Now, the Earth moves through absolute space, if for no other reason than its motion around the Sun; it moves in one direction in January, then in the opposite direction six months later, in June. Correspondingly, we on Earth should measure the speed of light to be different in different directions, and the differences should change with the seasons—though only very slightly (about 1 part in 10,000), because the Earth moves so slowly compared to light.

To verify this prediction was a fascinating challenge for experimental physicists. Albert Michelson, a twenty-eight-year-old American, took up the challenge in 1881, using an exquisitely accurate experimental technique (now called "Michelson interferometry"[2]) that he had invented. But try as he might, Michelson could find no evidence whatsoever for any variation of light speed with direction. The speed turned out to be the same in *all* directions and at *all* seasons in his initial 1881 experiments in Potsdam, Germany, and the same to much higher precision in later 1887 experiments that Michelson performed in Cleveland, Ohio, jointly with a chemist, Edward Morley. Michelson reacted with a mixture of elation at his discovery and dismay at its consequences. Heinrich Weber and most other physicists of the 1890s reacted with skepticism.

It was easy to be skeptical. Interesting experiments are often terribly difficult—so difficult, in fact, that regardless of how carefully they are carried out, they can give wrong results. Just one little abnormality in the apparatus, or one tiny uncontrolled fluctuation in its temperature, or one unexpected vibration of the floor beneath it, might alter the experiment's final result. Thus, it is not surprising that physicists of today, like physicists of the 1890s, are occasionally confronted by terribly difficult experiments which conflict with each other or conflict with our deeply cherished beliefs about the nature of the Universe and its physical laws. Recent examples are experiments that purported to discover a "fifth force" (one not present in the standard, highly successful physical laws) and other experiments denying that such a force exists; also experiments claiming to discover "cold fusion" (a phenomenon forbidden by the standard laws, if physicists understand those laws correctly) and other experiments denying that cold fusion occurs. Almost always the experiments that threaten our deeply cherished beliefs are wrong; their radical results are artifacts of experimental error. However, occasionally they are right and point the way toward a revolution in our understanding of nature.